```
# Name:Rutuja Barelikar
# Roll No: 10
# Assignment 5: Decision Tree Classification
#
================================================================================

import pandas as pd
from IPython.display import display

from sklearn.model_selection import train_test_split
from sklearn.tree import DecisionTreeClassifier
from sklearn.metrics import accuracy_score, precision_score
from sklearn.metrics import recall_score, f1_score, confusion_matrix

# Create Dataset
# ------------------------------------------------------------------------

data = {
   "Age": [22, 25, 28, 30, 32, 35, 38, 40, 42, 45,
         24, 27, 31, 36, 39, 44, 26, 33, 37, 41],

   "Salary": [20000, 25000, 30000, 35000, 40000,
            45000, 50000, 55000, 60000, 65000,
            22000, 28000, 36000, 48000, 52000,
            70000, 27000, 42000, 47000, 58000],

   "Online_Hours": [1, 2, 2, 3, 3,
                 4, 4, 5, 5, 6,
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                1, 2, 3, 4, 5,
                6, 2, 4, 5, 6],

    "Purchased": ["No", "No", "No", "No", "Yes",
                  "Yes", "Yes", "Yes", "Yes", "Yes",
                  "No", "No", "No", "Yes", "Yes",
                  "Yes", "No", "Yes", "Yes", "Yes"]
}

df = pd.DataFrame(data)

print("=" * 60)
print("STEP 1: CUSTOMER DATASET")
print("=" * 60)
display(df)

# Convert Target Variable
# ------------------------------------------------------------------------

df["Purchased"] = df["Purchased"].map({"No": 0, "Yes": 1})

print("\nDataset After Encoding Target Variable")
display(df)

#  Separate Features and Target
# ------------------------------------------------------------

X = df[["Age", "Salary", "Online_Hours"]]
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y = df["Purchased"]

# Split Dataset
# ------------------------------------------------------------

X_train, X_test, y_train, y_test = train_test_split(
    X,
    y,
    test_size=0.20,
    random_state=42
)

print("\nTraining Features")
display(X_train)

print("\nTesting Features")
display(X_test)

print("\nTraining Target")
display(y_train.to_frame())

print("\nTesting Target")
display(y_test.to_frame())

# Create Decision Tree Model
# ------------------------------------------------------------

model = DecisionTreeClassifier(max_depth=3, random_state=42)
```

```
# Train Model
model.fit(X_train, y_train)

# Prediction
# ------------------------------------------------------------

y_pred = model.predict(X_test)

print("\nActual Values")
print(y_test.values)

print("\nPredicted Values")
print(y_pred)

#Model Evaluation
# ------------------------------------------------------------

accuracy = accuracy_score(y_test, y_pred)
precision = precision_score(y_test, y_pred, zero_division=0)
recall = recall_score(y_test, y_pred, zero_division=0)
f1 = f1_score(y_test, y_pred, zero_division=0)

print("\nModel Performance")
print("-" * 30)
print("Accuracy  :", round(accuracy * 100, 2), "%")
print("Precision :", round(precision, 2))
print("Recall    :", round(recall, 2))
```

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print("F1 Score  :", round(f1, 2))

# Confusion Matrix
# ------------------------------------------------------------

print("\nConfusion Matrix")
print(confusion_matrix(y_test, y_pred))

#  Predict New Customer
# ------------------------------------------------------------

new_customer = pd.DataFrame({
   "Age": [34],
   "Salary": [45000],
   "Online_Hours": [4]
})

prediction = model.predict(new_customer)

print("\nNew Customer Details")
display(new_customer)

if prediction[0] == 1:
   print("Prediction: Customer Will Purchase")
else:
   print("Prediction: Customer Will Not Purchase")

print("Decision Tree Classification Completed Successfully!")
```

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STEP 1: CUSTOMER DATASET

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	Age	Salary	Online_Hours	Purchased
0	22	20000	1	No
1	25	25000	2	No
2	28	30000	2	No
3	30	35000	3	No
4	32	40000	3	Yes
5	35	45000	4	Yes
6	38	50000	4	Yes
7	40	55000	5	Yes
8	42	60000	5	Yes
9	45	65000	6	Yes
10	24	22000	1	No
11	27	28000	2	No
12	31	36000	3	No
13	36	48000	4	Yes
14	39	52000	5	Yes
15	44	70000	6	Yes
16	26	27000	2	No

	Age	Salary	Online_Hours	Purchased
17	33	42000	4	Yes
18	37	47000	5	Yes
19	41	58000	6	Yes

Dataset After Encoding Target Variable

	Age	Salary	Online_Hours	Purchased
0	22	20000	1	0
1	25	25000	2	0
2	28	30000	2	0
3	30	35000	3	0
4	32	40000	3	1
5	35	45000	4	1
6	38	50000	4	1
7	40	55000	5	1
8	42	60000	5	1
9	45	65000	6	1
10	24	22000	1	0
11	27	28000	2	0
12	31	36000	3	0
13	36	48000	4	1
14	39	52000	5	1

	Age	Salary	Online_Hours	Purchased
15	44	70000	6	1
16	26	27000	2	0
17	33	42000	4	1
18	37	47000	5	1
19	41	58000	6	1

Training Features

	Age	Salary	Online_Hours
8	42	60000	5
5	35	45000	4
11	27	28000	2
3	30	35000	3
18	37	47000	5
16	26	27000	2
13	36	48000	4
2	28	30000	2
9	45	65000	6
19	41	58000	6
4	32	40000	3
12	31	36000	3
7	40	55000	5

	Age	Salary	Online_Hours
10	24	22000	1
14	39	52000	5
6	38	50000	4

Testing Features

	Age	Salary	Online_Hours
0	22	20000	1
17	33	42000	4
15	44	70000	6
1	25	25000	2

Training Target

	Purchased
8	1
5	1
11	0
3	0
18	1
16	0
13	1
2	0

	Purchased
9	1
19	1
4	1
12	0
7	1
10	0
14	1
6	1

Testing Target

	Purchased
0	0
17	1
15	1
1	0

Actual Values

[0 1 1 0]

Predicted Values

[0 1 1 0]

Model Performance

Accuracy : 100.0 %

Precision : 1.0

Recall : 1.0

F1 Score : 1.0

Confusion Matrix

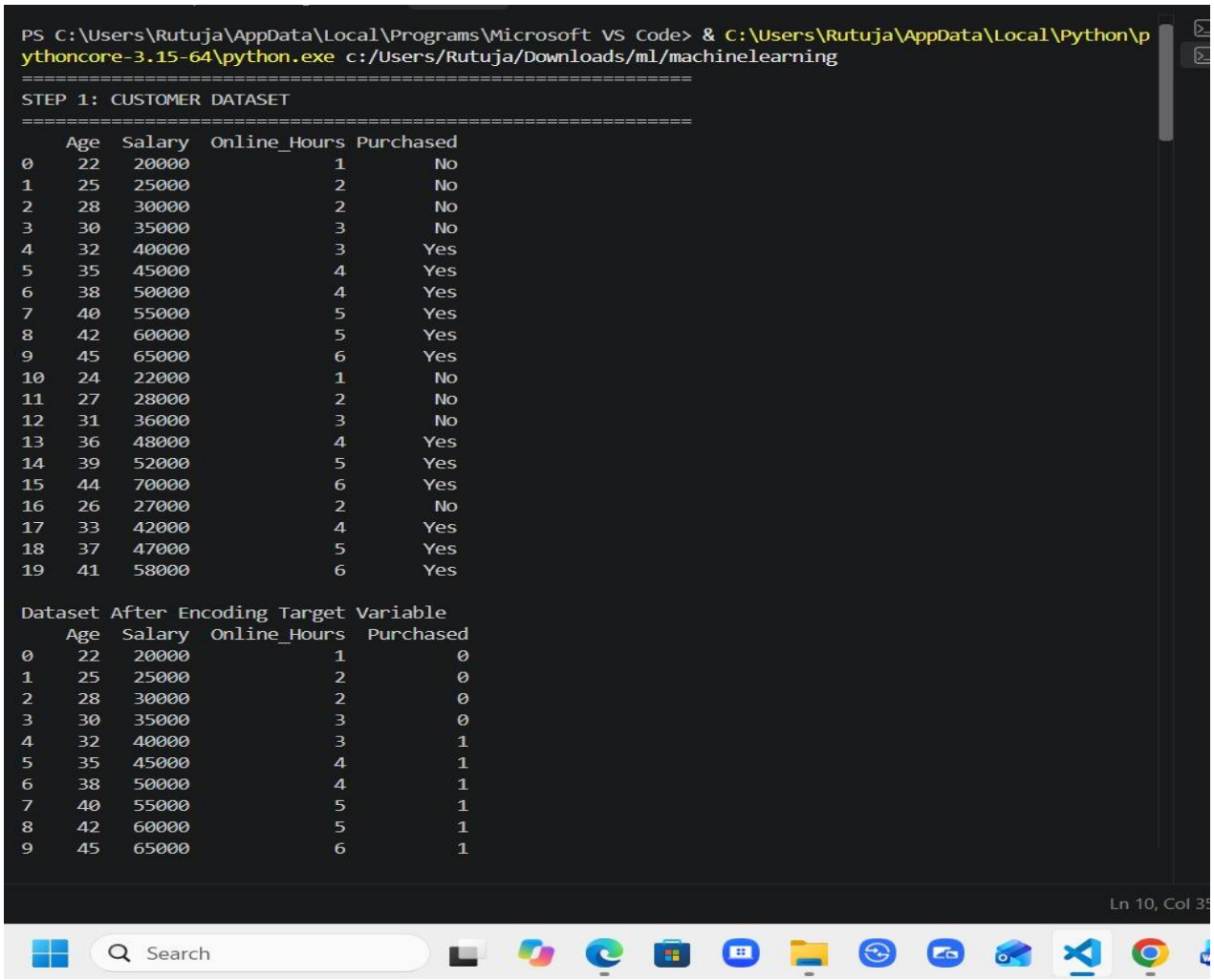
[[2 0]

[0 2]]

New Customer Details

	Age	Salary	Online_Hours
0	34	45000	4

Prediction: Customer Will Purchase



```
3    30   35000              3         No
4    32   40000              3        Yes
5    35   45000              4        Yes
6    38   50000              4        Yes
7    40   55000              5        Yes
8    42   60000              5        Yes
9    45   65000              6        Yes
10   24   22000              1         No
11   27   28000              2         No
12   31   36000              3         No
13   36   48000              4        Yes
14   39   52000              5        Yes
15   44   70000              6        Yes
16   26   27000              2         No
17   33   42000              4        Yes
18   37   47000              5        Yes
19   41   58000              6        Yes

Dataset After Encoding Target Variable
    Age  Salary  Online_Hours  Purchased
0    22   20000             1          0
1    25   25000             2          0
2    28   30000             2          0
3    30   35000             3          0
4    32   40000             3          1
5    35   45000             4          1
6    38   50000             4          1
7    40   55000             5          1
8    42   60000             5          1
9    45   65000             6          1
10   24   22000             1          0
11   27   28000             2          0
12   31   36000             3          0
13   36   48000             4          1
14   39   52000             5          1
15   44   70000             6          1
16   26   27000             2          0
17   33   42000             4          1
18   37   47000             5          1
```

```
Testing Target
    Purchased
0           0
17          1
15          1
1           0

Actual Values
[0 1 1 0]

Predicted Values
[0 1 1 0]

Model Performance
------------------------------
Accuracy  : 100.0 %
Precision : 1.0
Recall    : 1.0
F1 Score  : 1.0

Confusion Matrix
[[2 0]
 [0 2]]

New Customer Details
   Age  Salary  Online_Hours
0   34   45000             4
Prediction: Customer Will Purchase
Decision Tree Classification Completed Successfully!
PS C:\Users\Rutuja\AppData\Local\Programs\Microsoft VS Code> & C:\Users\Rutuja\AppData\Local\Python\p
ythoncore-3.15-64\python.exe c:/Users/Rutuja/Downloads/ml/machinelearning
============================================================
STEP 1: CUSTOMER DATASET
============================================================
    Age  Salary  Online_Hours Purchased
0    22   20000             1        No
1    25   25000             2        No
2    28   30000             2        No
```

```
17   33   42000              4        Yes
18   37   47000              5        Yes
19   41   58000              6        Yes

Dataset After Encoding Target Variable
    Age  Salary  Online_Hours  Purchased
0    22   20000             1          0
1    25   25000             2          0
2    28   30000             2          0
3    30   35000             3          0
4    32   40000             3          1
5    35   45000             4          1
6    38   50000             4          1
7    40   55000             5          1
8    42   60000             5          1
9    45   65000             6          1
10   24   22000             1          0
11   27   28000             2          0
12   31   36000             3          0
13   36   48000             4          1
14   39   52000             5          1
15   44   70000             6          1
16   26   27000             2          0
17   33   42000             4          1
18   37   47000             5          1
19   41   58000             6          1

Training Features
    Age  Salary  Online_Hours
8    42   60000             5
5    35   45000             4
11   27   28000             2
3    30   35000             3
18   37   47000             5
16   26   27000             2
13   36   48000             4
2    28   30000             2
9    45   65000             6
19   41   58000             6
```

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```
  3    30   35000              3          0
  4    32   40000              3          1
  5    35   45000              4          1
  6    38   50000              4          1
  7    40   55000              5          1
  8    42   60000              5          1
  9    45   65000              6          1
  10   24   22000              1          0
  11   27   28000              2          0
  12   31   36000              3          0
  13   36   48000              4          1
  14   39   52000              5          1
  15   44   70000              6          1
  16   26   27000              2          0
  17   33   42000              4          1
  18   37   47000              5          1
  19   41   58000              6          1

  Training Features
      Age  Salary  Online_Hours
  8    42   60000             5
  5    35   45000             4
  11   27   28000             2
  3    30   35000             3
  18   37   47000             5
  16   26   27000             2
  13   36   48000             4
  2    28   30000             2
  9    45   65000             6
  19   41   58000             6
  4    32   40000             3
  12   31   36000             3
  7    40   55000             5
  10   24   22000             1
  14   39   52000             5
  6    38   50000             4

  Testing Features
      Age  Salary  Online_Hours
```